Benchmark Charts

Harrington's Insights: benchmark charts, standard compressors against the author's originals, modified tools and mutants. Assembled 29 October 2026 from recorded results; one run per tool per file, and every result was decompressed and checked byte for byte when it was recorded.

Classes. Standard: the stock tool. Original: the author's own designs (HCM, hcmx, Harrington Exchange). Modified: a tuned encoder whose output the stock decoder reads unchanged (bzip2-hi, xz-hi). Mutant: an existing compressor whose decoder was changed (lpaq1-h, the Exchange->Blosc2 hybrid, Exchange->xz).

Sets follow Andrew Goodwin's review: the Canterbury corpus, the first 1 MB of each Silesia file, and his synthetic set; plus the author's biased packed-data set and six radio captures from earlier runs.

How to read the dot charts. Each dot is one tool's total over a set. Up and down is size (lower is smaller); left to right is speed in MB of original data per second on a log scale (further right is faster). The dashed line joins the tools that nothing else beats on both size and speed; dots with a dark ring sit on it. Our tools are drawn larger with bold labels. The line charts show each file separately.

Data sources. All three of Goodwin's sets were run fresh, one run of all 14 tools per file on the same machine (29 Oct 2026), so speeds compare directly. Peak memory is the highest of compression and decode across the set's files.

Timing notes. Times are wall clock including program start-up, so on small files the fast tools all look about the same; sizes are exact. lpaq1-h's compression time is the sum of its six tries (it keeps the best). ZPAQ sizes have its 1,015-byte per-file archive header removed, as in Goodwin's review. On the biased and radio sets decode times are earlier in-process measurements and compression times were mostly not recorded, so those two sets have decode charts only; ZPAQ's radio decode time is an estimate.

RESULTS

Biased packed data (4 files) (2,173,577 bytes). Smallest overall: order-0 rANS. Our best: Harrington Exchange at 1,011,727 bytes (decode 25.1 ms).

Radio SDR captures (6 files) (3,752,512 bytes). Smallest overall: ZPAQ -m5 (decode estimated). Our best: xz-hi at 2,299,596 bytes (decode 171.0 ms).

Biased file 12bit_noise (524,288 bytes): our best Harrington Exchange 324,507; best fast standard tool (decode under 0.1 s) order-0 rANS 323,335; smallest of all paq8px -8 323,153. The hybrid beats Blosc2 on both size (324,765 vs 360,045) and decode (0.69 vs 0.76 ms).

Biased file biased_bits (125,000 bytes): our best Harrington Exchange 19,722; best fast standard tool (decode under 0.1 s) order-0 rANS 17,925; smallest of all order-0 rANS 17,925.

Biased file biased_bytes (1,000,000 bytes): our best Harrington Exchange 321,128; best fast standard tool (decode under 0.1 s) order-0 rANS 290,572; smallest of all paq8px -8 290,460.

Biased file 5bit_skewed (524,289 bytes): our best Exchange->Blosc2 hybrid 345,725; best fast standard tool (decode under 0.1 s) order-0 rANS 340,229; smallest of all order-0 rANS 340,229.

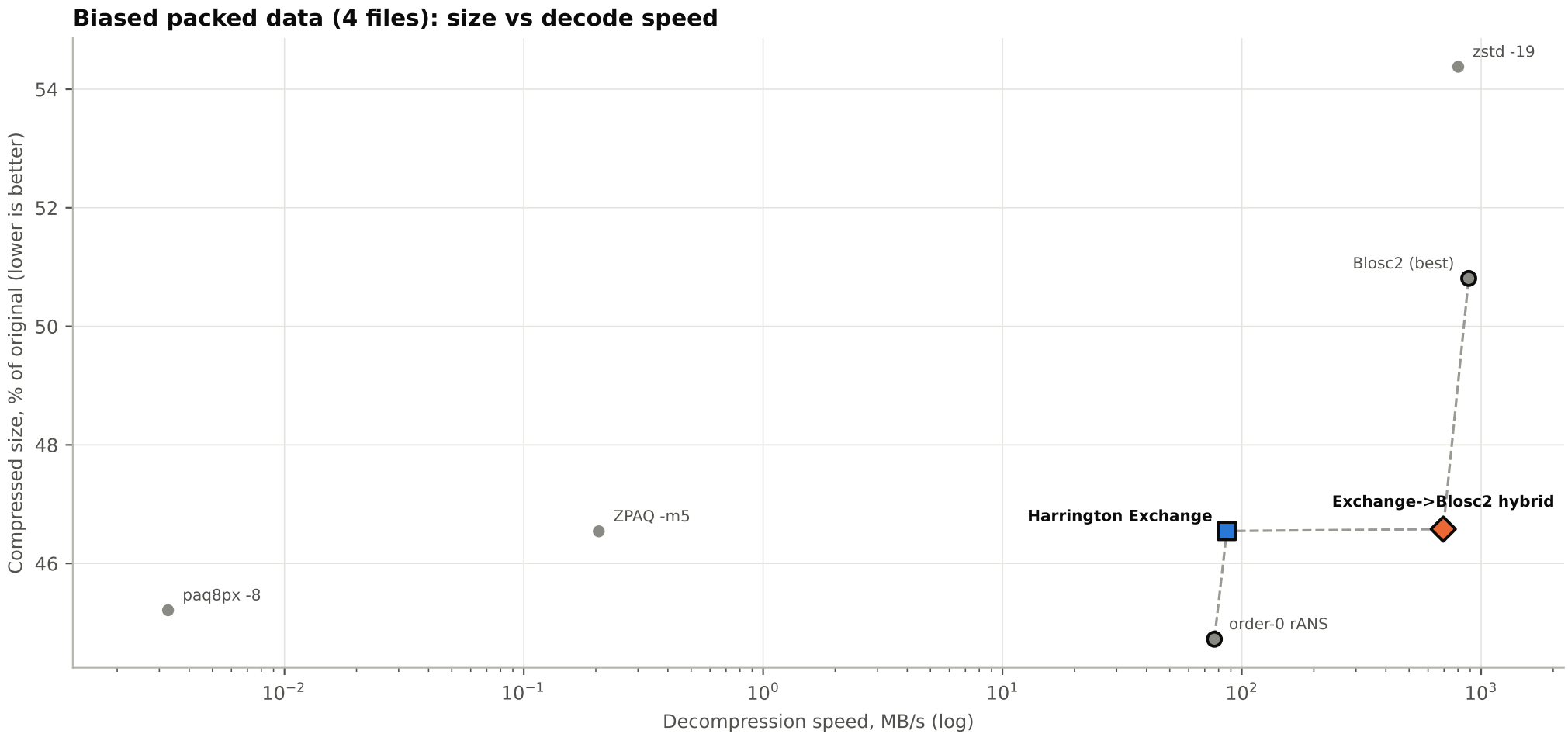
Benchmark Charts (continued)

Biased set note: the order-0 rANS coder was smaller than the Exchange on all four files (972,061 vs 1,011,727 bytes in total) but decodes about 9x slower than the hybrid; paq8px is smallest on three files but takes minutes to decode. On the 5-bit file the Exchange and the hybrid beat paq8px outright.

Canterbury (2,810,784 bytes, 14 tools). Smallest overall: HCM v3.4 max at 350,326 bytes. Best standard: ZPAQ -m5 at 350,542. Our best: HCM v3.4 max at 350,326 bytes (-0.06% vs best standard). Ours on the decode-speed trade-off line: HCM v3.4 max, hcmx fast, HCM v3.4 fast, xz-hi. Smaller AND faster to decode: xz-hi beats bzip2 -9, xz -9e; HCM v3.4 fast beats lpaq1 -6; HCM v3.4 max beats lpaq1 -6, ZPAQ -m5; hcmx fast beats lpaq1 -6.

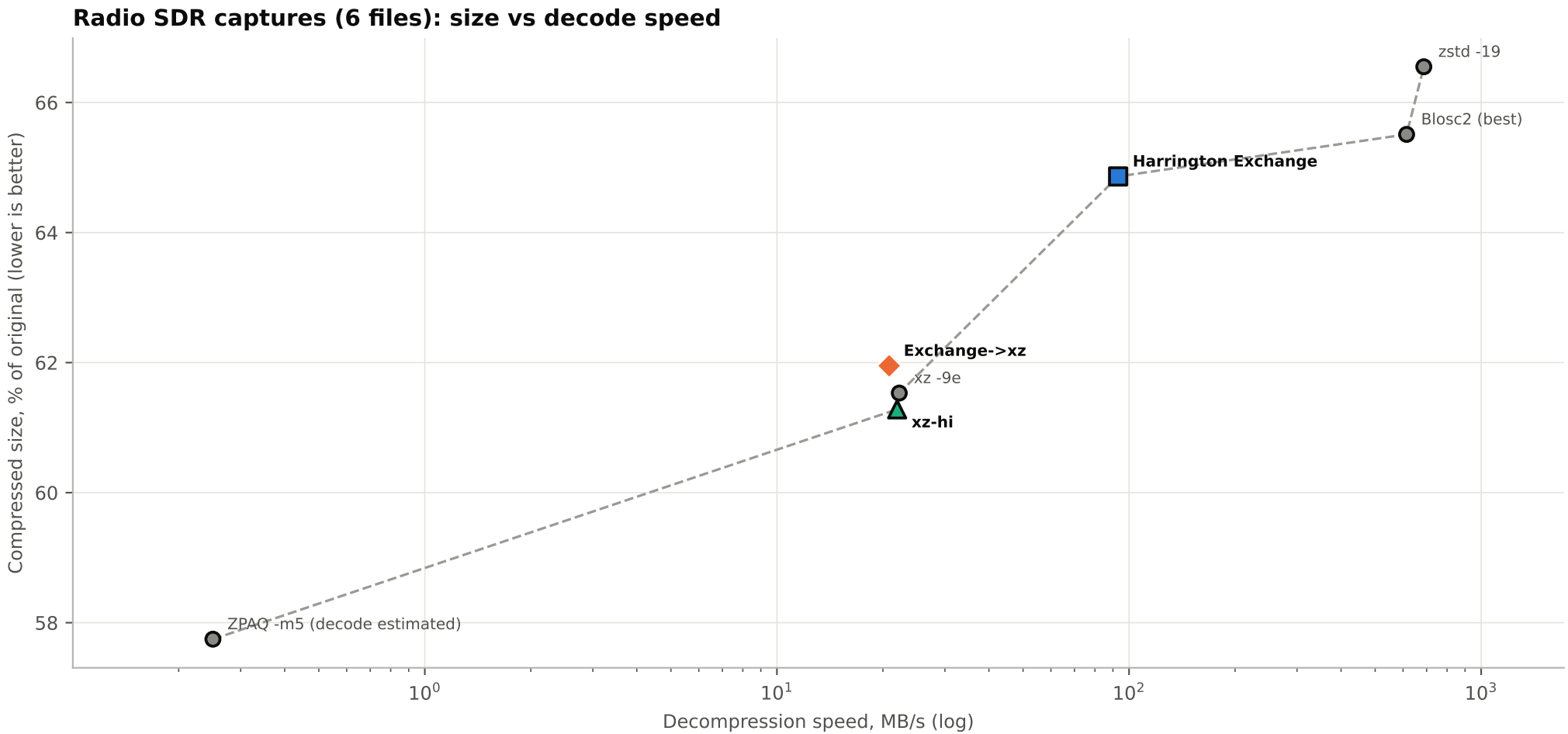
Silesia 1 MB sample (12,582,912 bytes, 14 tools). Smallest overall: ZPAQ -m5 at 3,418,919 bytes. Best standard: ZPAQ -m5 at 3,418,919. Our best: HCM v3.4 max at 3,544,604 bytes (+3.68% vs best standard). Ours on the decode-speed trade-off line: HCM v3.4 max, HCM v3.4 fast, xz-hi. Smaller AND faster to decode: bzip2-hi beats bzip2 -9; xz-hi beats bzip2 -9, xz -9e; HCM v3.4 fast beats lpaq1 -6.

Synthetic set (17,999,934 bytes, 14 tools). Smallest overall: lpaq1-h at 8,113,745 bytes. Best standard: ZPAQ -m5 at 8,201,855. Our best: lpaq1-h at 8,113,745 bytes (-1.07% vs best standard). Ours on the decode-speed trade-off line: lpaq1-h, HCM v3.4 max, hcmx fast, HCM v3.4 fast, xz-hi. Smaller AND faster to decode: xz-hi beats bzip2 -9; HCM v3.4 fast beats lpaq1 -6, ZPAQ -m5; HCM v3.4 max beats lpaq1 -6, ZPAQ -m5; hcmx fast beats lpaq1 -6, ZPAQ -m5; lpaq1-h beats ZPAQ -m5.



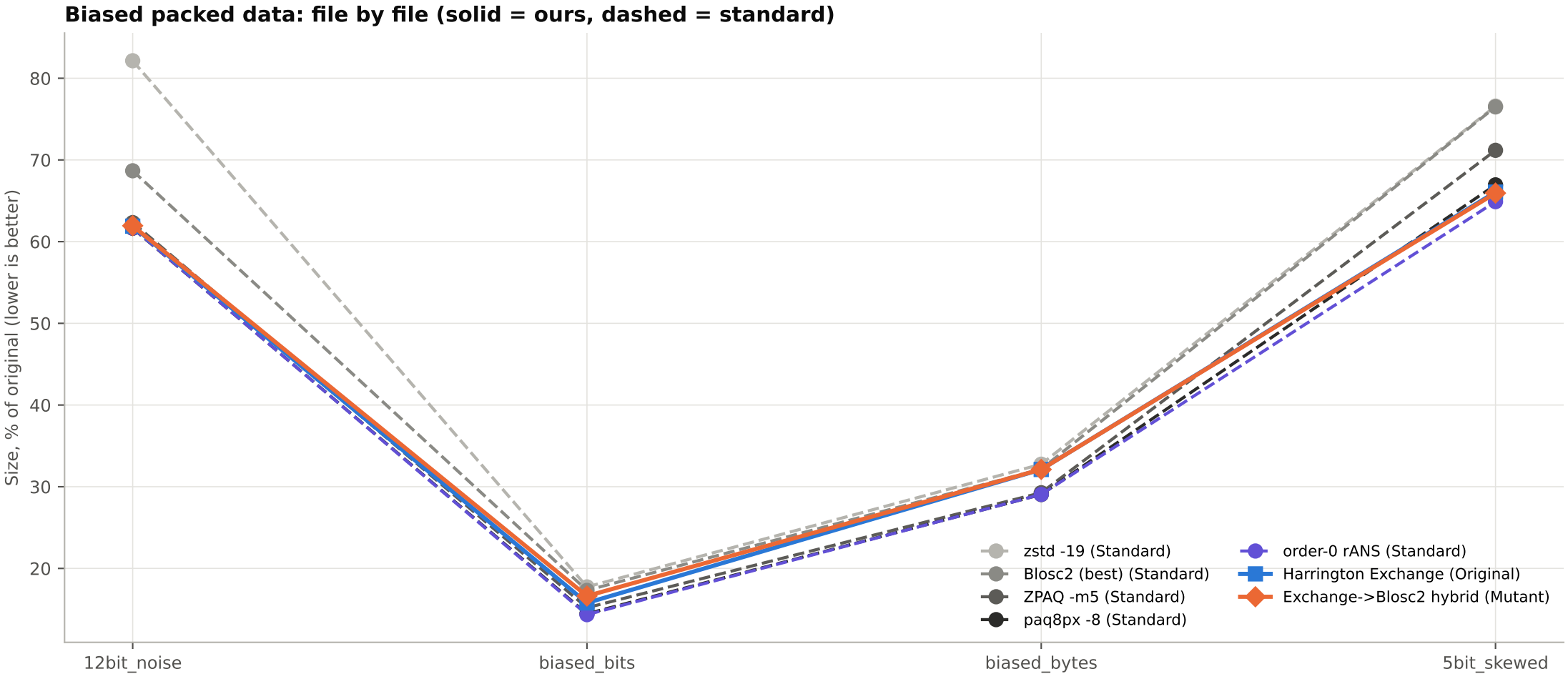
Compression times were not recorded for this set; decode times are earlier in-process measurements. Size only (no decode time recorded): Brotli 11 51.5%; 7-Zip ultra 53.2%; xz -9e 53.3%; OpenZL 55.9%.

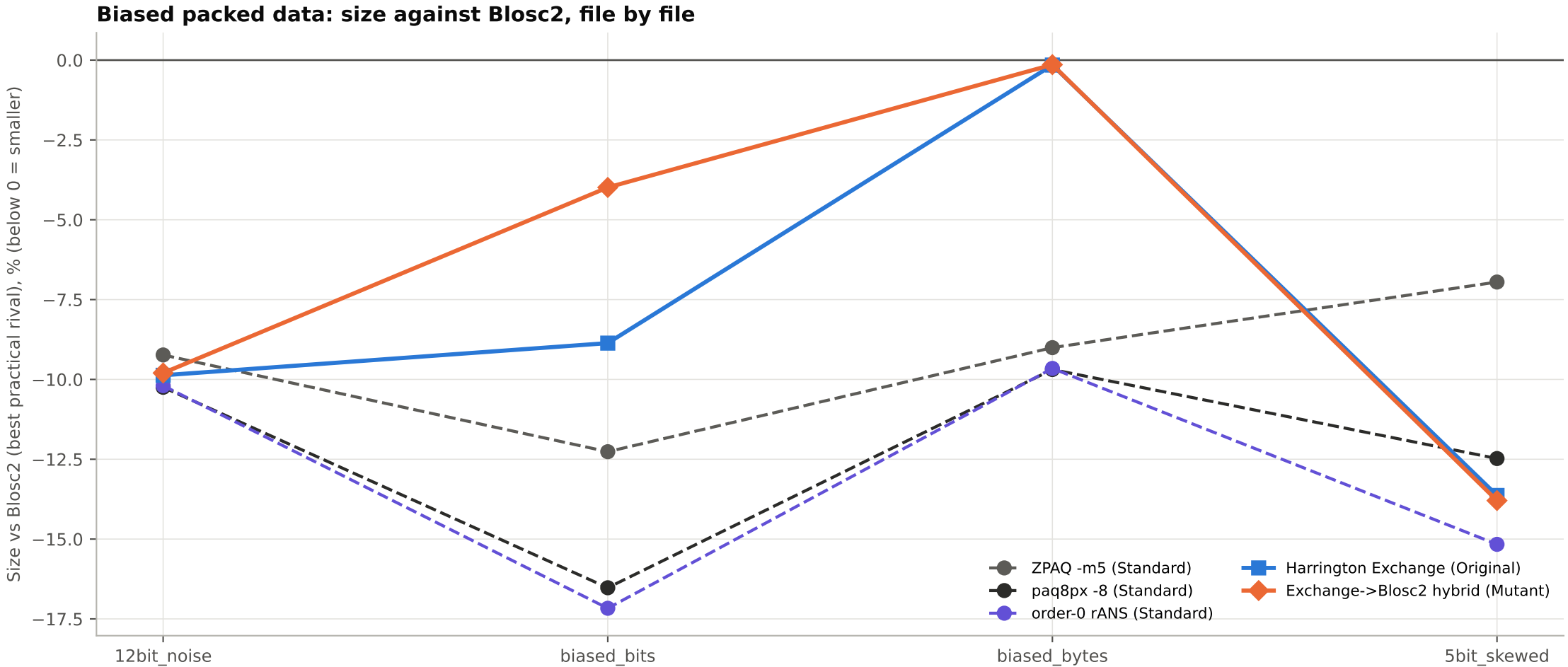
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

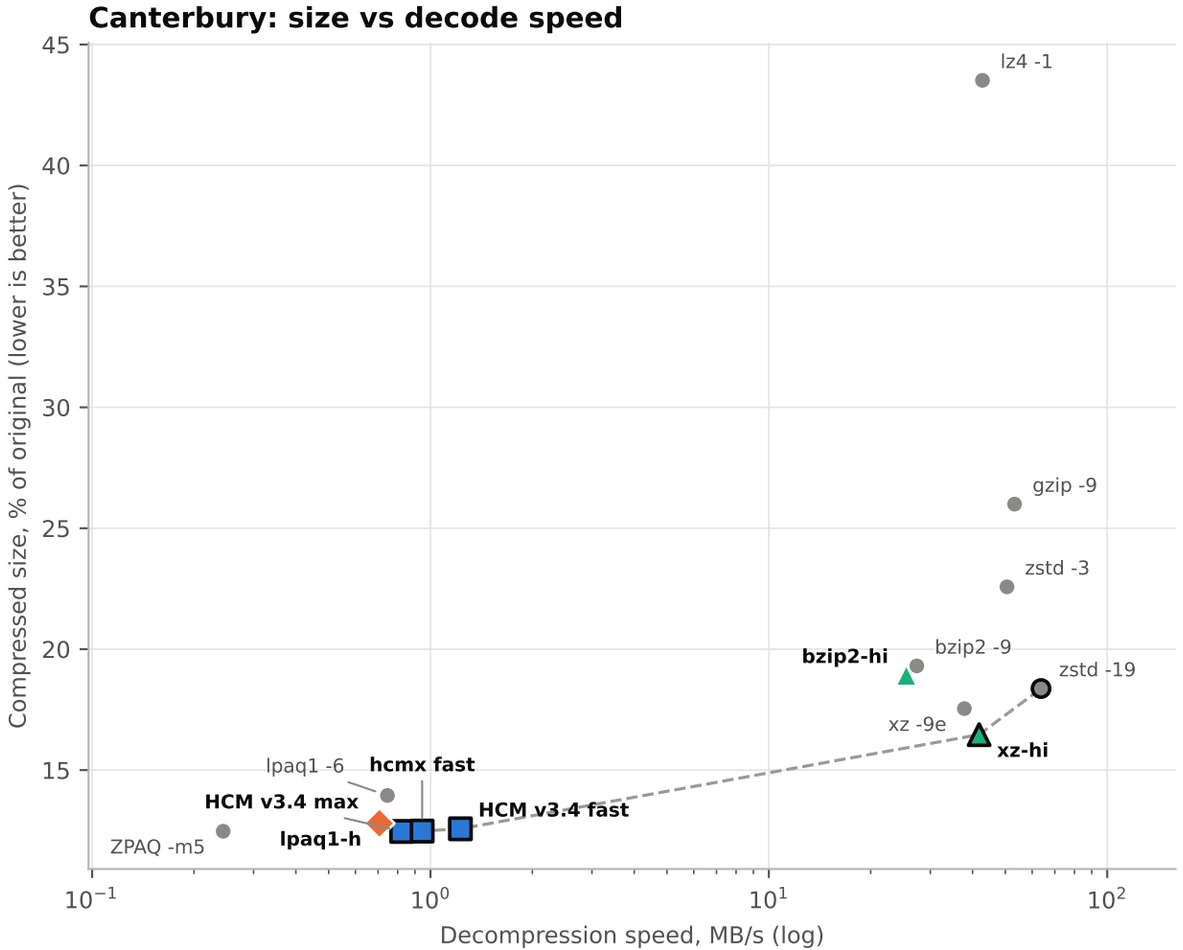


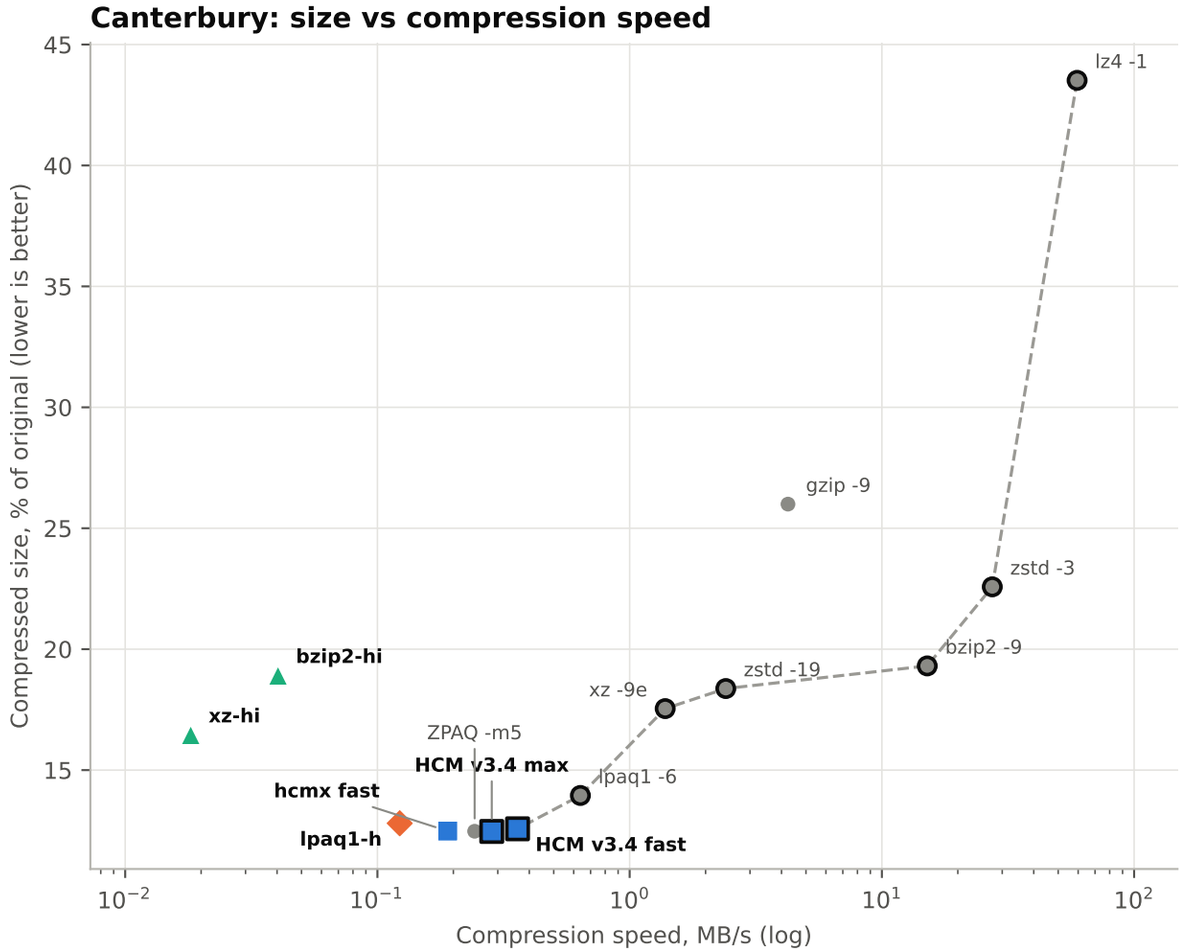
Compression times were not recorded for this set; decode times are earlier in-process measurements. Size only (no decode time recorded): order-0 rANS 70.7%.

Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

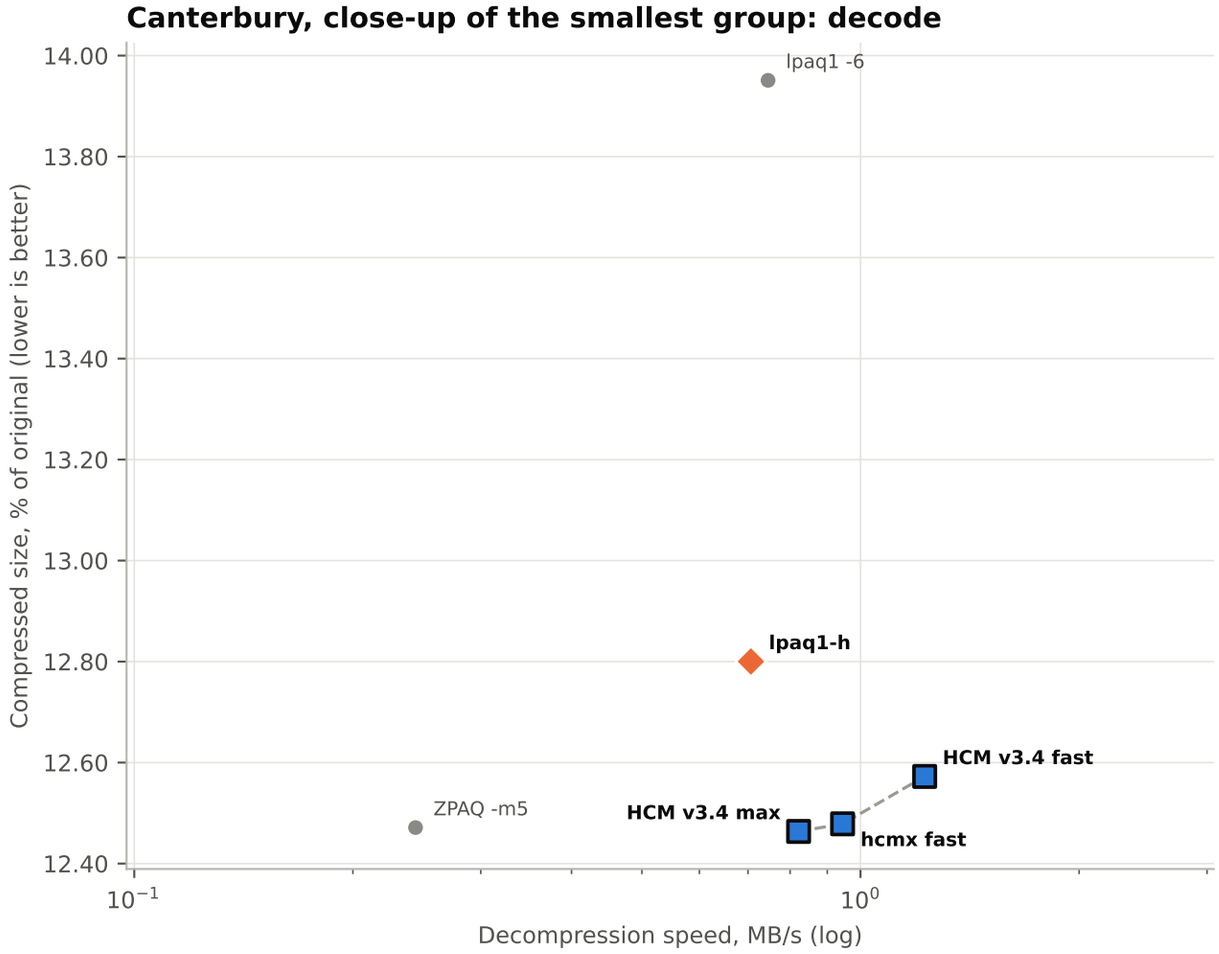

Biased packed data: file by file (solid = ours, dashed = standard)
Size, % of original (lower is better)
80
70
60
50
40
30
20
12bit_noise
biased_bits
biased_bytes
5bit_skewed
zstd -19 (Standard)
Blosc2 (best) (Standard)
ZPAQ -m5 (Standard)
paq8px -8 (Standard)
order-0 rANS (Standard)
Harrington Exchange (Original)
Exchange->Blosc2 hybrid (Mutant)

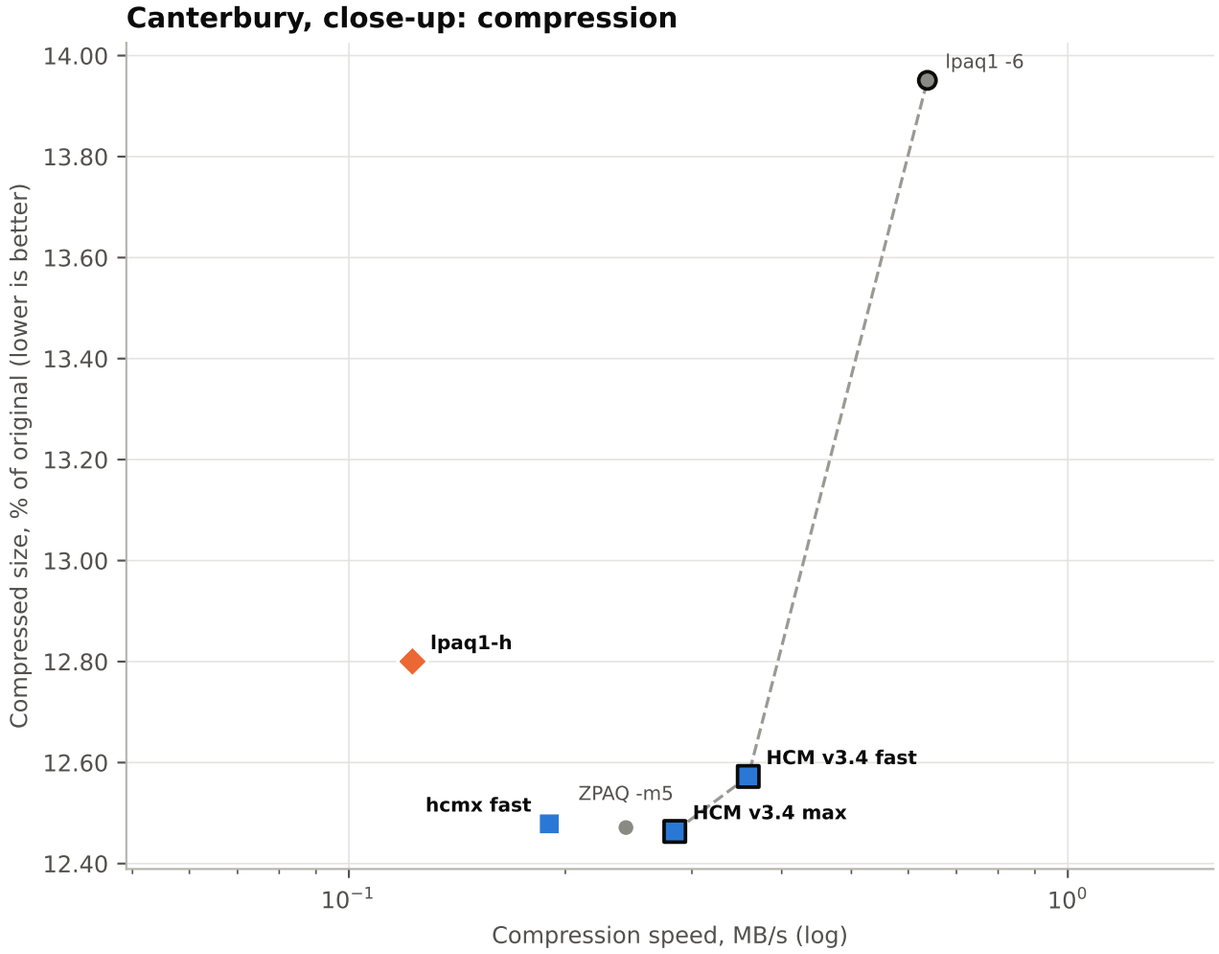

Biased packed data: size against Blosc2, file by file
Size vs Blosc2 (best practical rival), % (below 0 = smaller)
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
12bit_noise
biased_bits
biased_bytes
5bit_skewed
ZPAQ -m5 (Standard)
paq8px -8 (Standard)
order-0 rANS (Standard)
Harrington Exchange (Original)
Exchange->Blosc2 hybrid (Mutant)

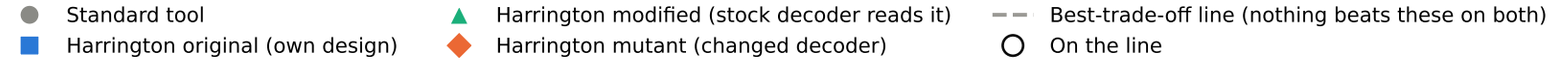


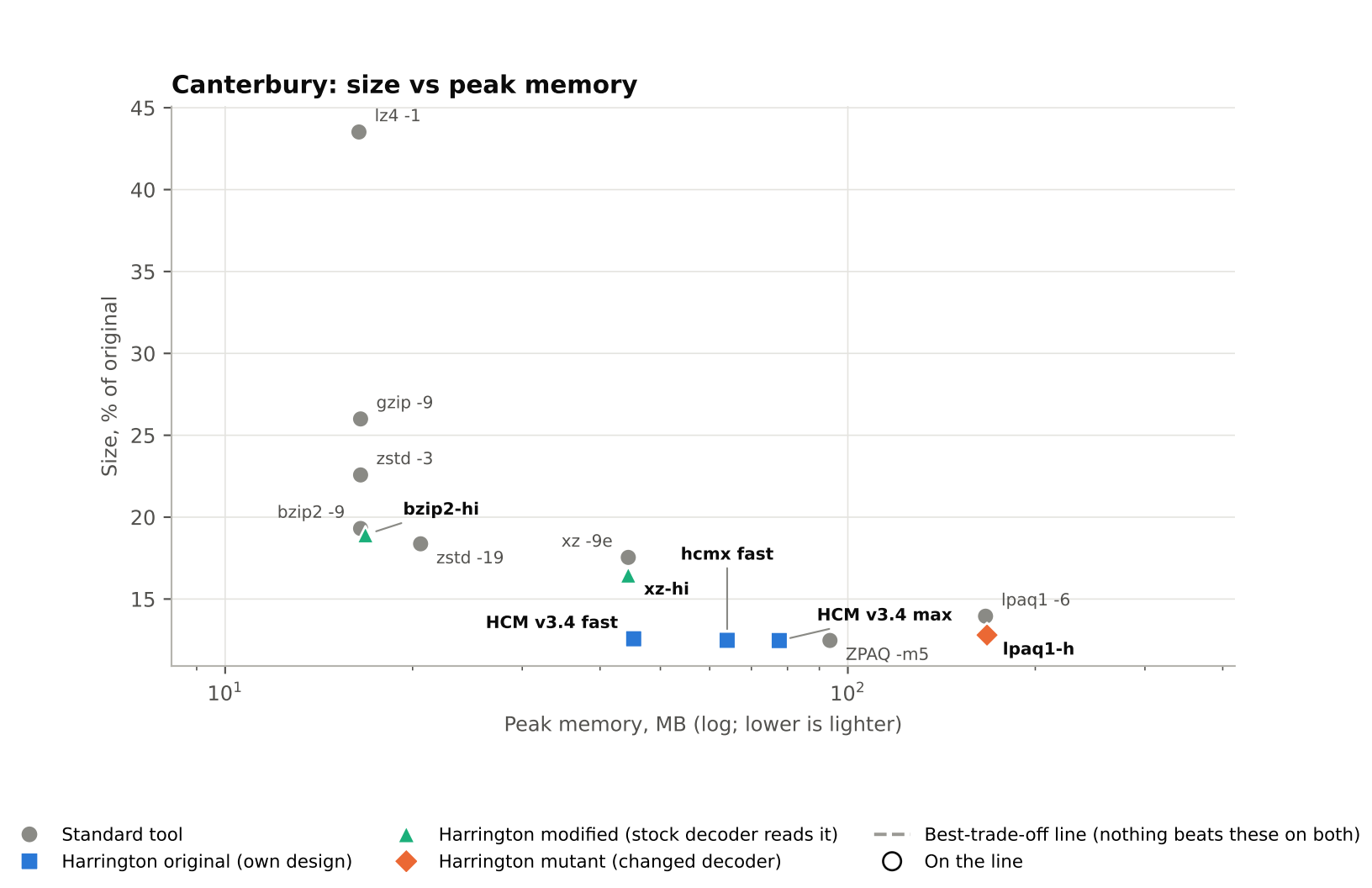


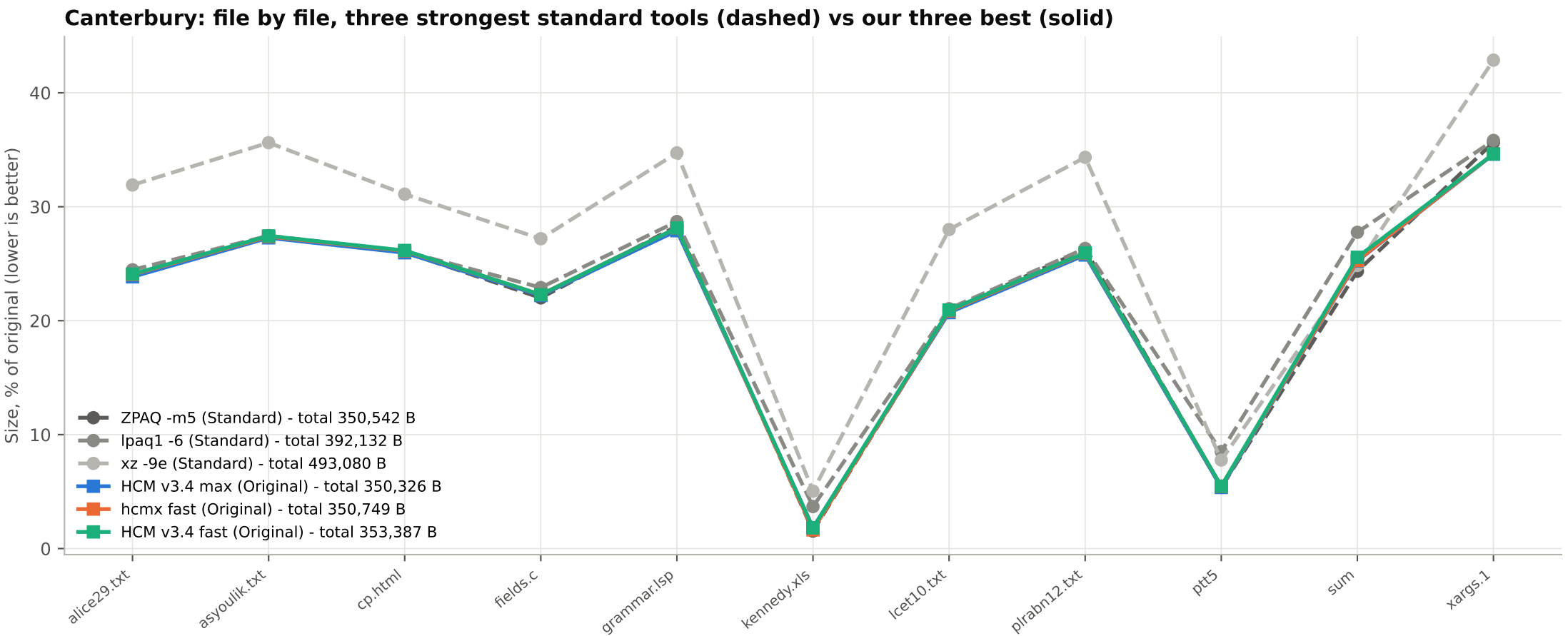
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

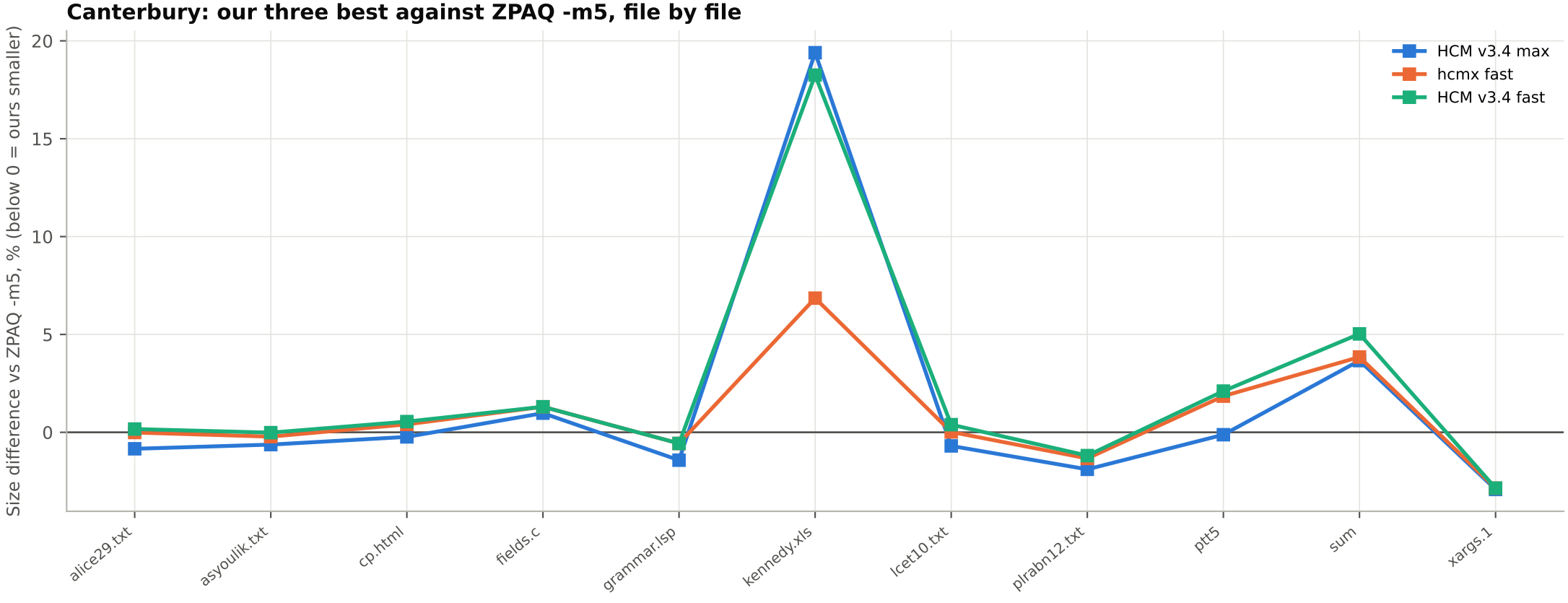

Canterbury, close-up of the smallest group: decode
Compressed size, % of original (lower is better)
Decompression speed, MB/s (log)
lpaq1 -6
lpaq1-h
HCM v3.4 fast
ZPAQ -m5
HCM v3.4 max
hcmx fast

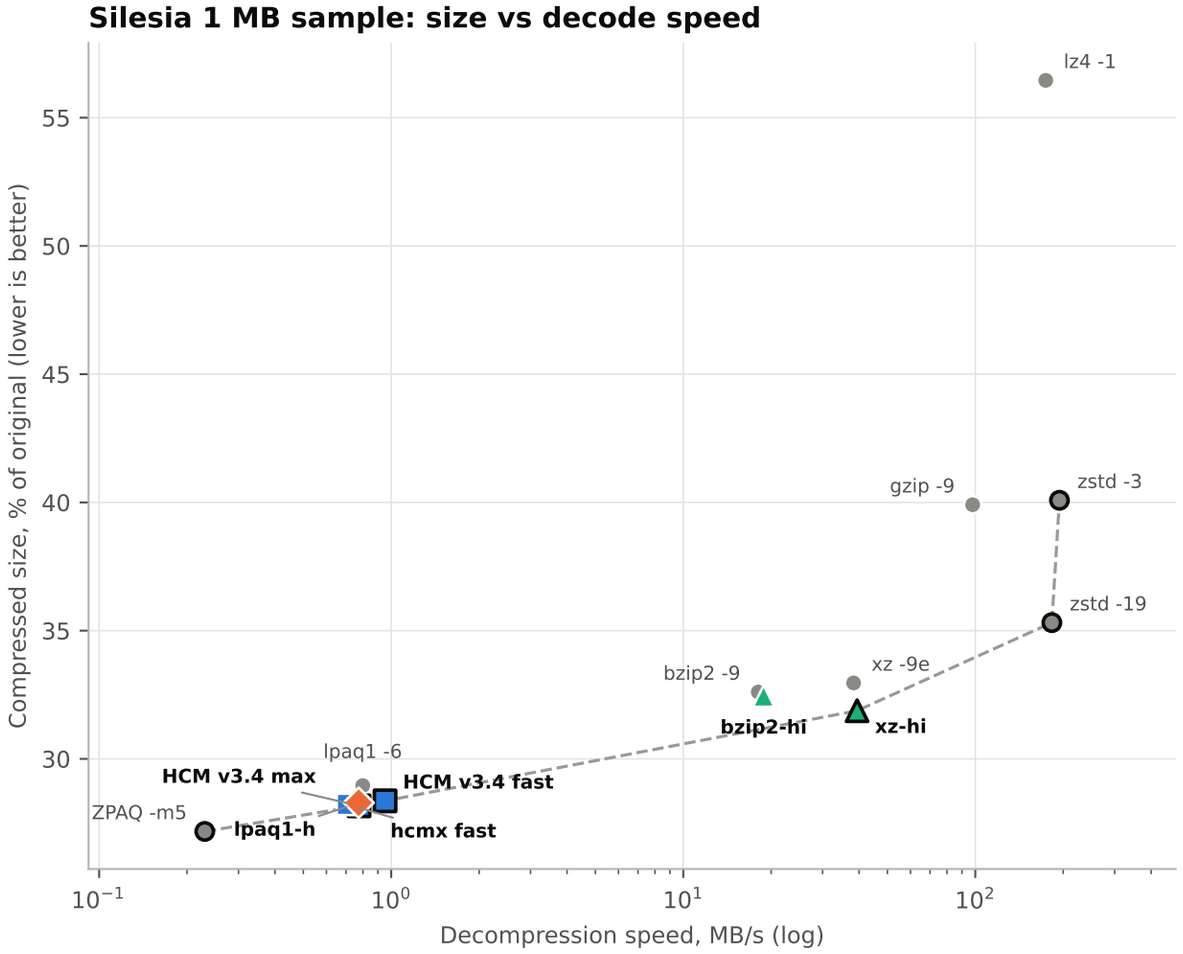

Canterbury, close-up: compression
Compressed size, % of original (lower is better)
Compression speed, MB/s (log)
lpaq1 -6
lpaq1-h
HCM v3.4 fast
hcmx fast
ZPAQ -m5
HCM v3.4 max

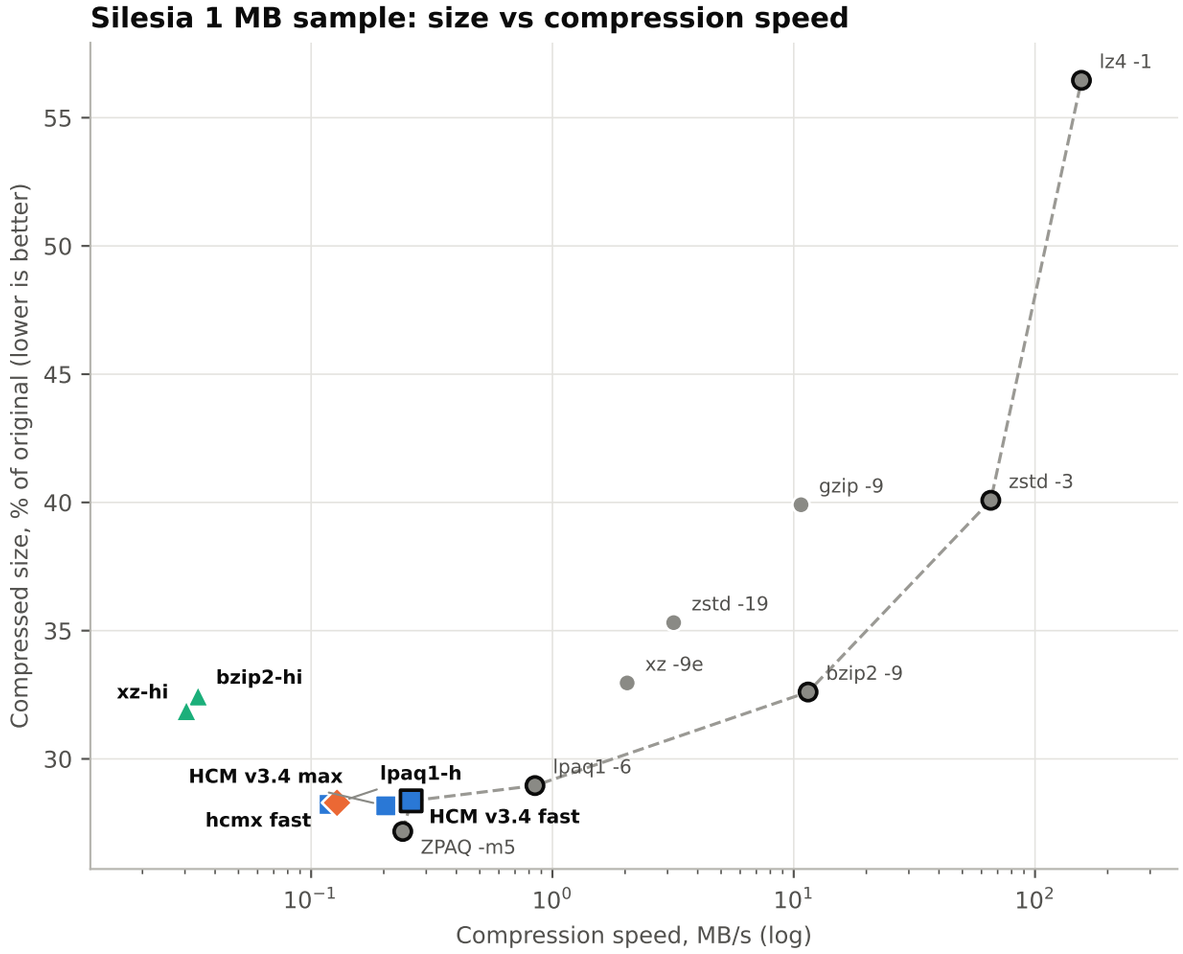

Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line


Canterbury: size vs peak memory
Size, % of original
Peak memory, MB (log; lower is lighter)
45
40
35
30
25
20
15
10^1
10^2
lz4 -1
gzip -9
zstd -3
bzip2 -9
bzip2-hi
zstd -19
xz -9e
hcmx fast
xz-hi
HCM v3.4 fast
HCM v3.4 max
ZPAQ -m5
lpaq1 -6
lpaq1-h
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

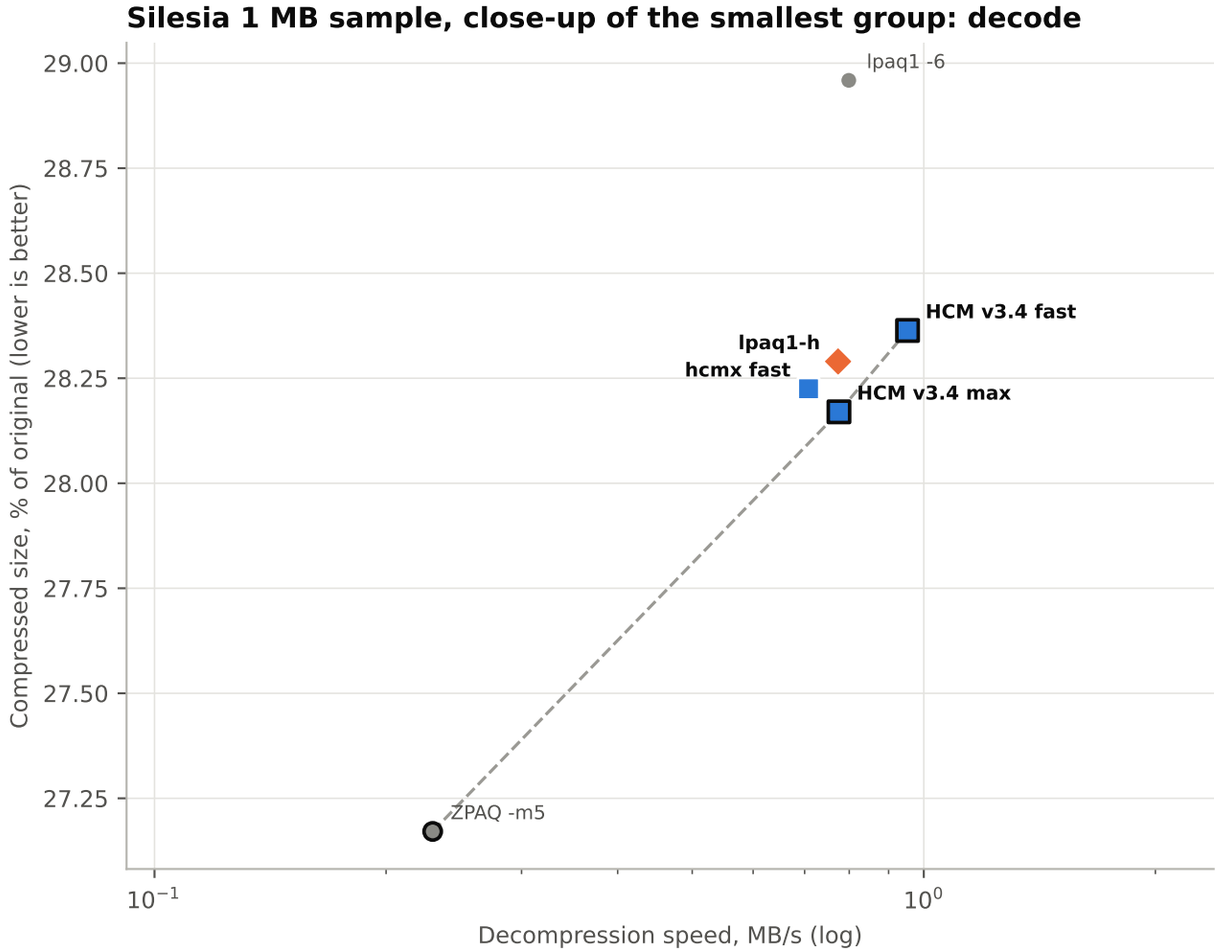

Canterbury: file by file, three strongest standard tools (dashed) vs our three best (solid)
Size, % of original (lower is better)
0
10
20
30
40
ZPAQ -m5 (Standard) - total 350,542 B
lpaq1 -6 (Standard) - total 392,132 B
xz -9e (Standard) - total 493,080 B
HCM v3.4 max (Original) - total 350,326 B
hcmx fast (Original) - total 350,749 B
HCM v3.4 fast (Original) - total 353,387 B
alice29.txt
asyoulik.txt
cp.html
fields.c
grammar.lsp
kennedy.xls
lcet10.txt
plrabn12.txt
ptt5
sum
xargs.1

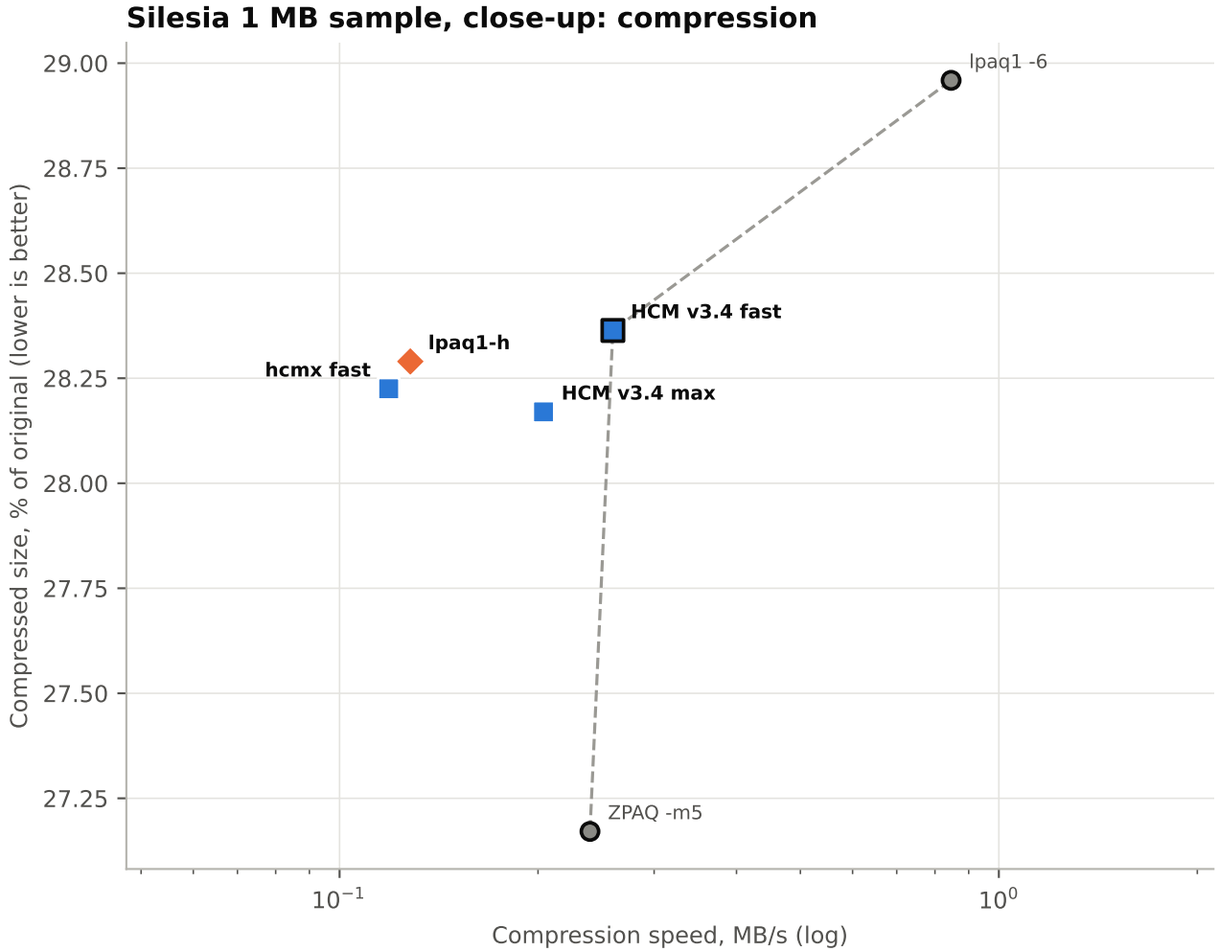

Canterbury: our three best against ZPAQ -m5, file by file
Size difference vs ZPAQ -m5, % (below 0 = ours smaller)
20
15
10
5
0
HCM v3.4 max
hcmx fast
HCM v3.4 fast
alice29.txt
asyoulik.txt
cp.html
fields.c
grammar.lsp
kennedy.xls
lcet10.txt
plrabn12.txt
ptt5
sum
xargs.1



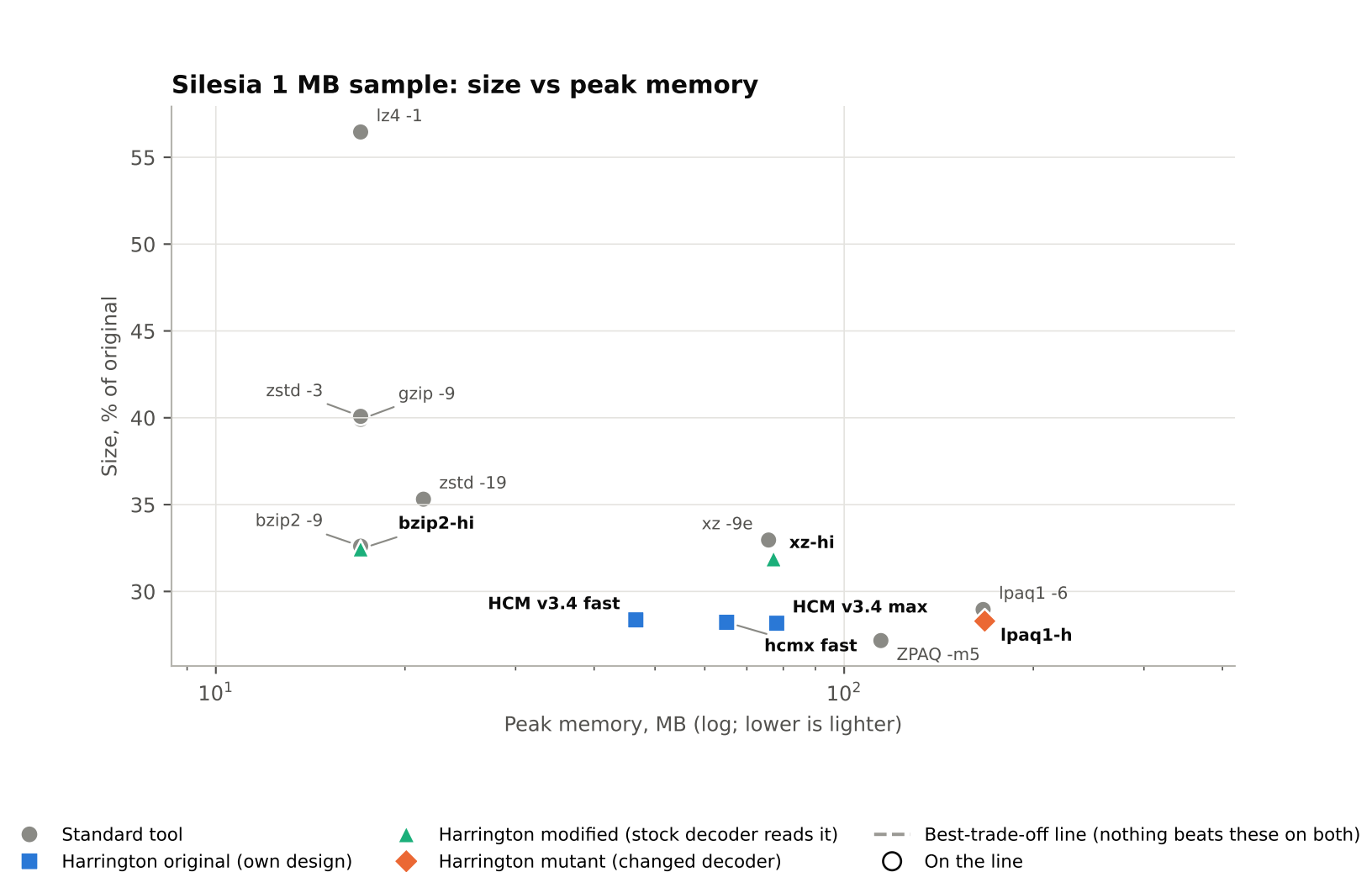


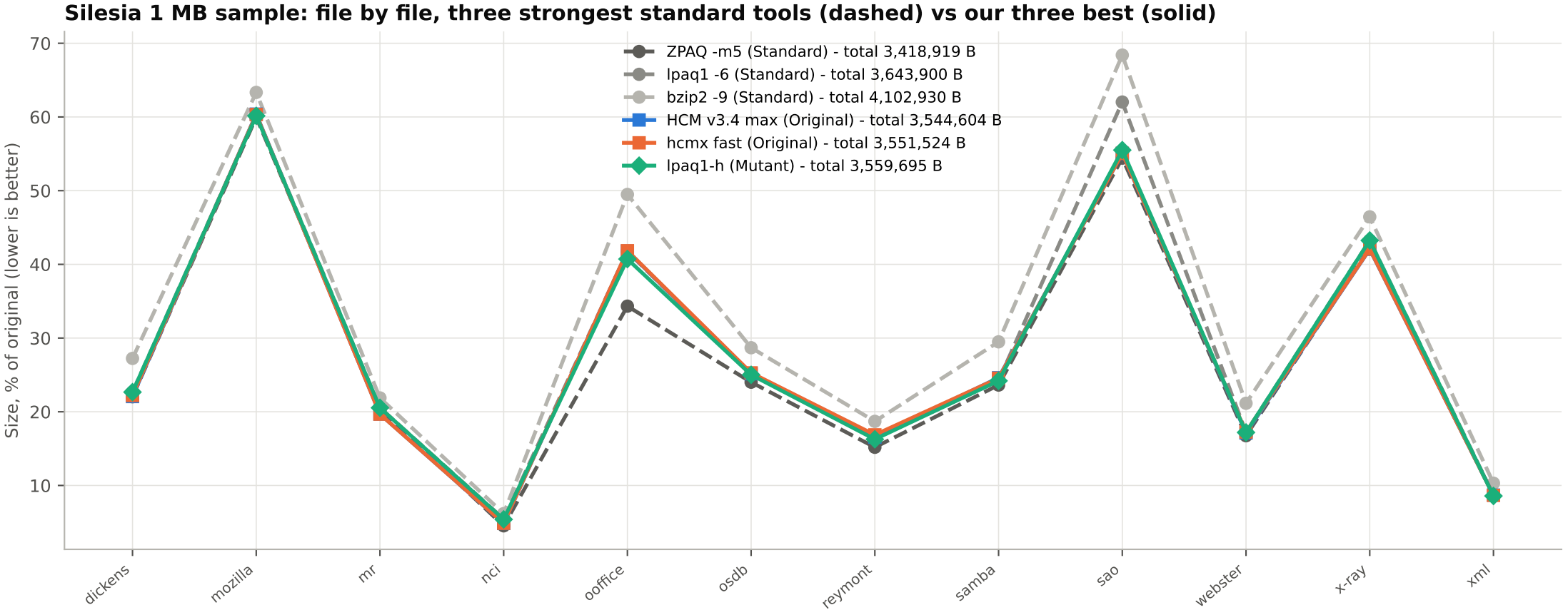
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

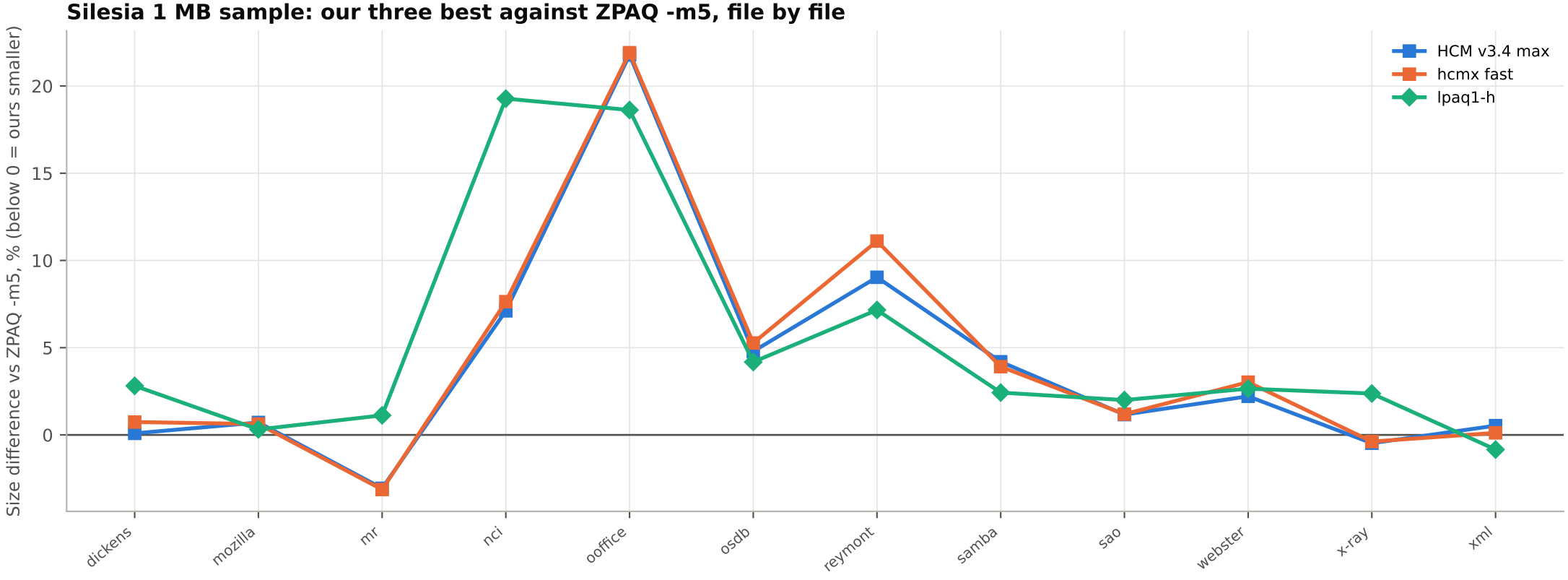


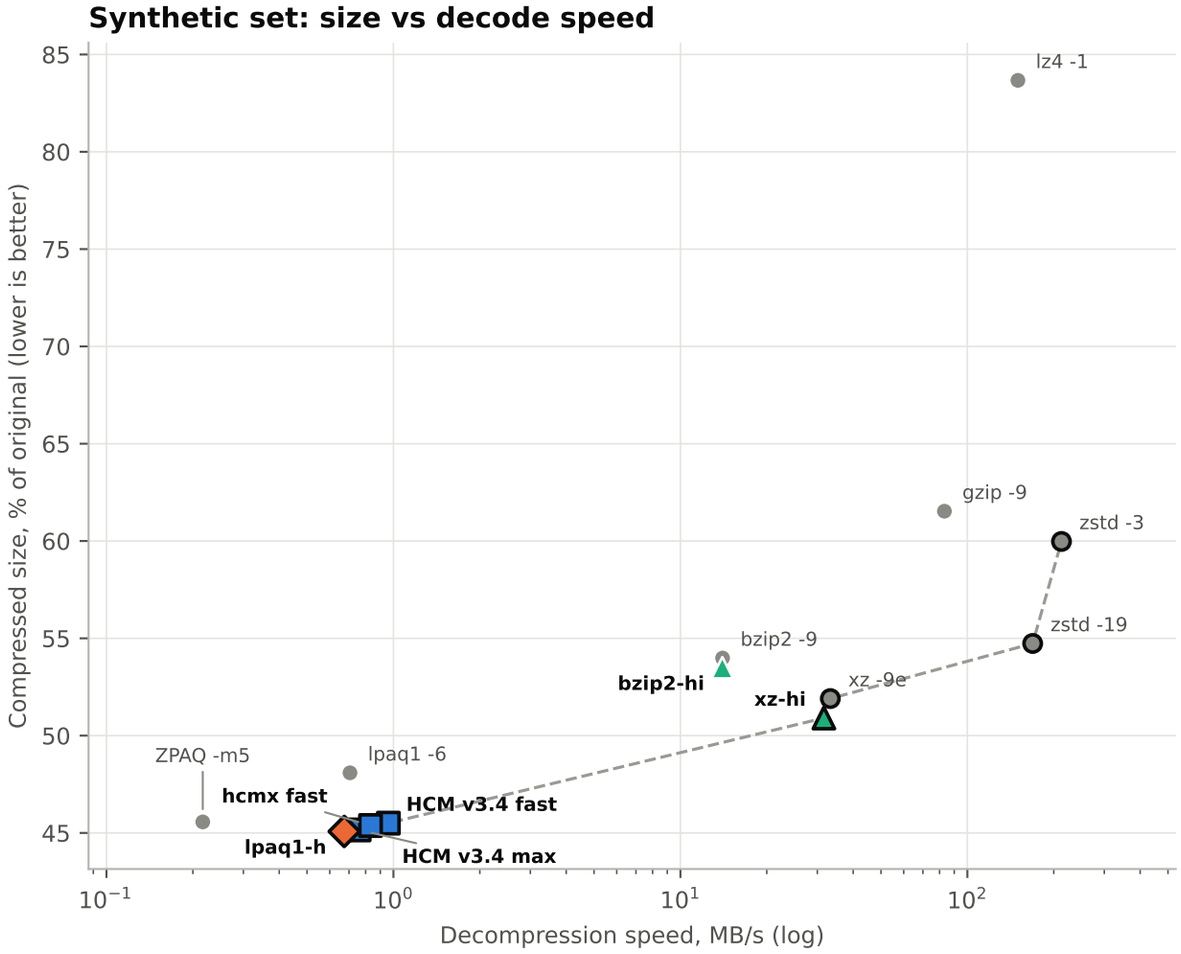


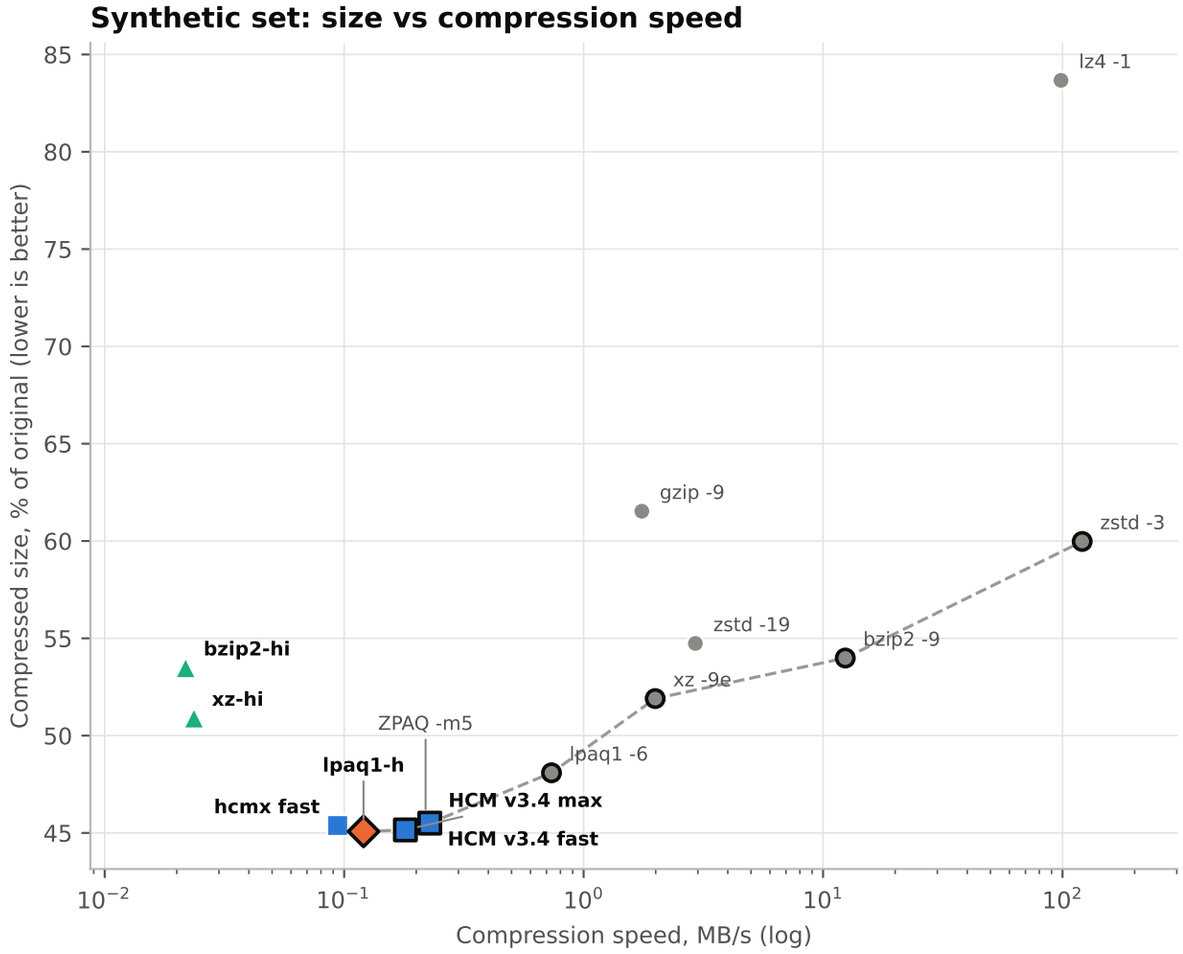
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

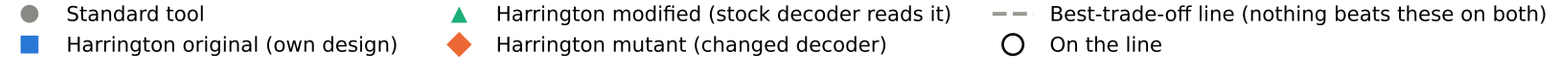

Silesia 1 MB sample: size vs peak memory
lz4 -1
zstd -3
gzip -9
zstd -19
bzip2 -9
bzip2-hi
xz -9e
xz-hi
HCM v3.4 fast
HCM v3.4 max
lpaq1 -6
lpaq1-h
hcmx fast
ZPAQ -m5
55
50
45
40
35
30
Size, % of original
10^1
10^2
Peak memory, MB (log; lower is lighter)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

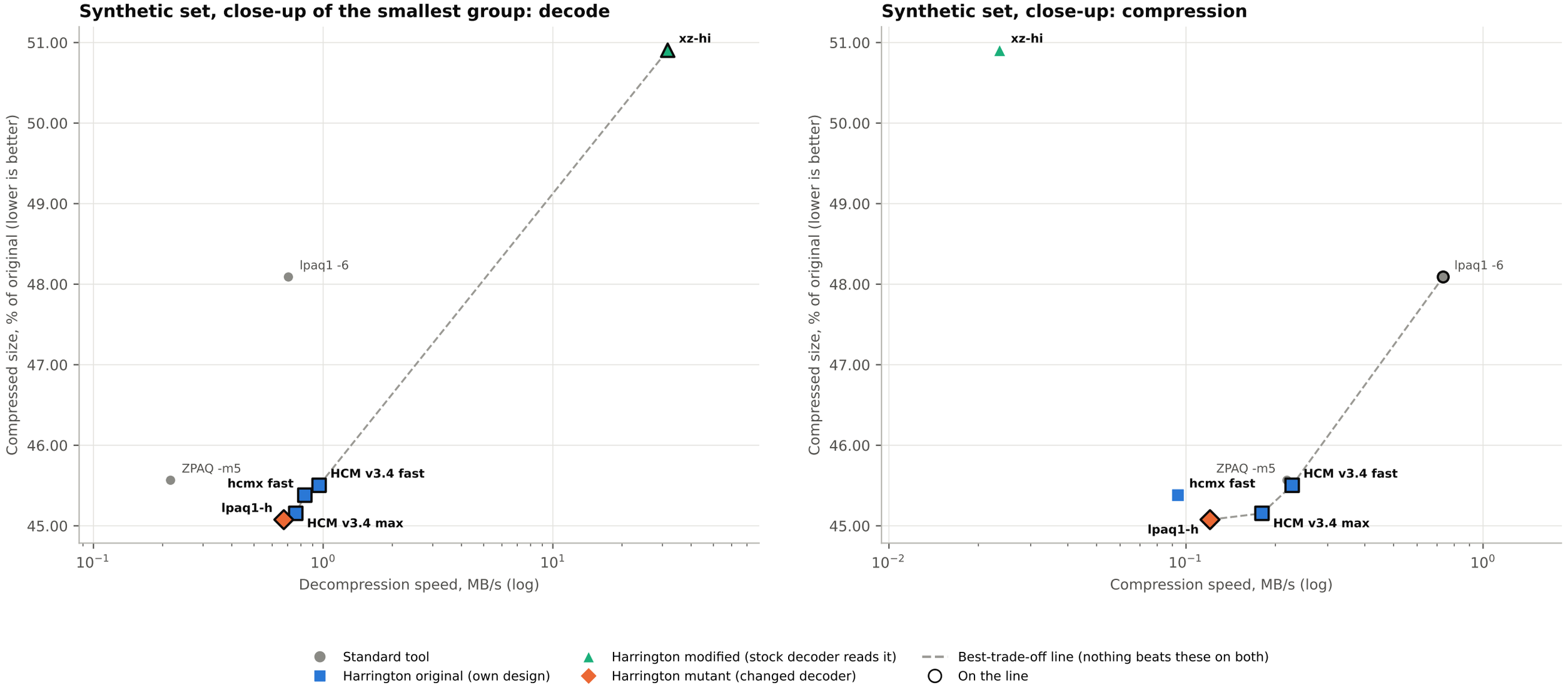

Silesia 1 MB sample: file by file, three strongest standard tools (dashed) vs our three best (solid)
Size, % of original (lower is better)
70
60
50
40
30
20
10
ZPAQ -m5 (Standard) - total 3,418,919 B
lpaq1 -6 (Standard) - total 3,643,900 B
bzip2 -9 (Standard) - total 4,102,930 B
HCM v3.4 max (Original) - total 3,544,604 B
hcmx fast (Original) - total 3,551,524 B
lpaq1-h (Mutant) - total 3,559,695 B
dickens
mozilla
mr
nci
ooffice
osdb
reymont
samba
sao
webster
x-ray
xml

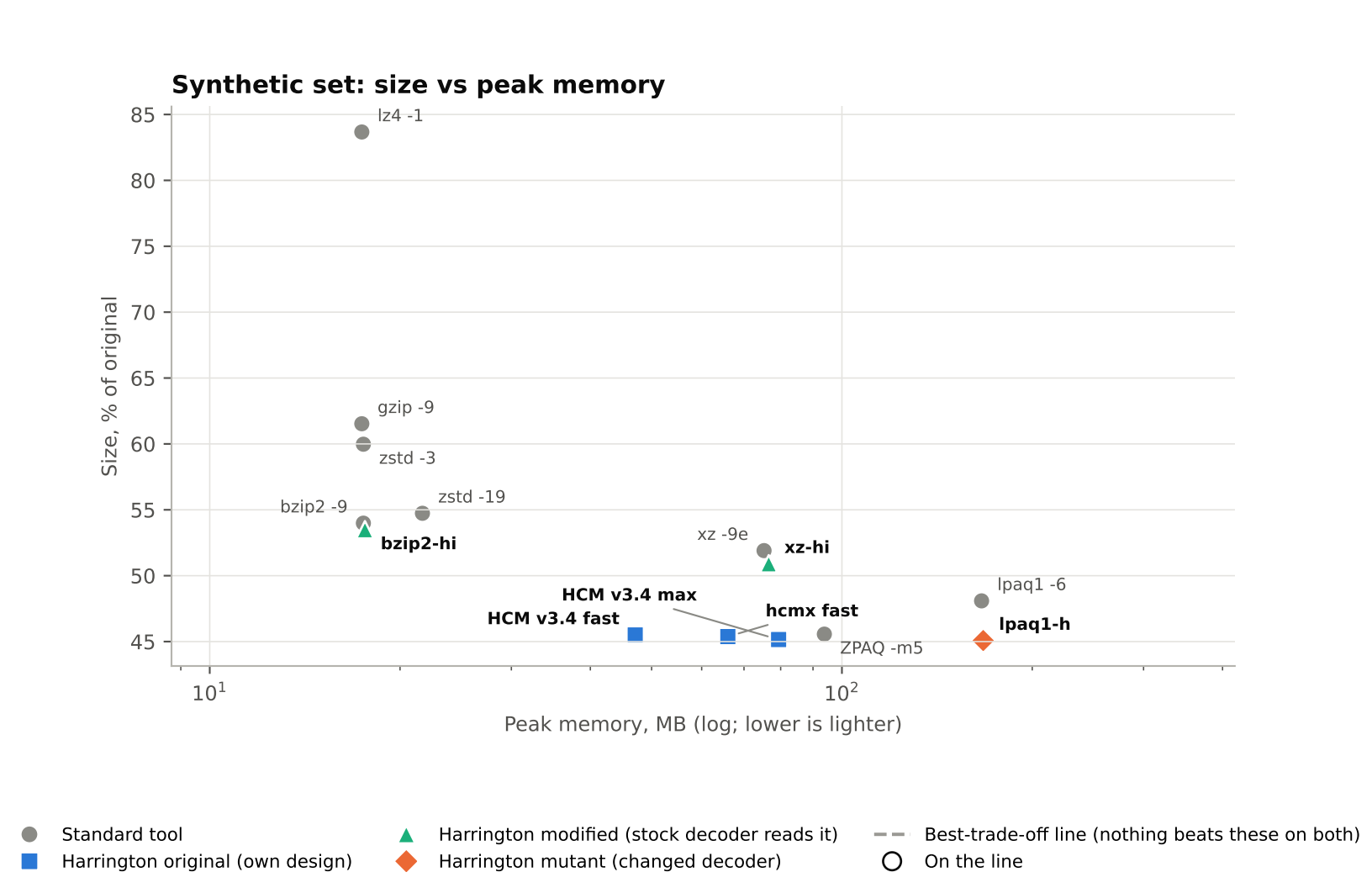

Silesia 1 MB sample: our three best against ZPAQ -m5, file by file
Size difference vs ZPAQ -m5, % (below 0 = ours smaller)
20
15
10
5
0
HCM v3.4 max
hcmx fast
lpaq1-h
dickens
mozilla
mr
nci
ooffice
osdb
reymont
samba
sao
webster
x-ray
xml

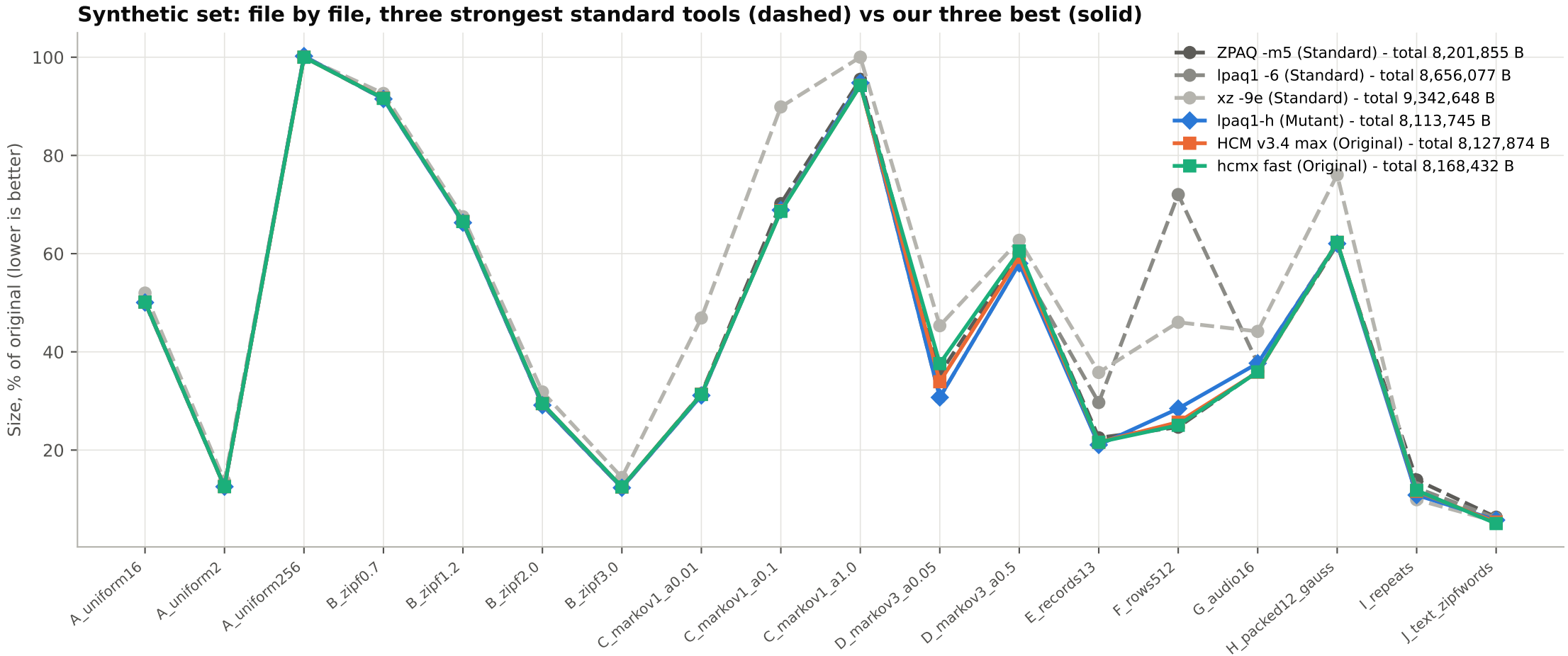

Synthetic set: size vs decode speed
Compressed size, % of original (lower is better)
Decompression speed, MB/s (log)
lz4 -1
gzip -9
zstd -3
zstd -19
bzip2 -9
xz -9e
bzip2-hi
xz-hi
ZPAQ -m5
lpaq1 -6
hcmx fast
HCM v3.4 fast
lpaq1-h
HCM v3.4 max

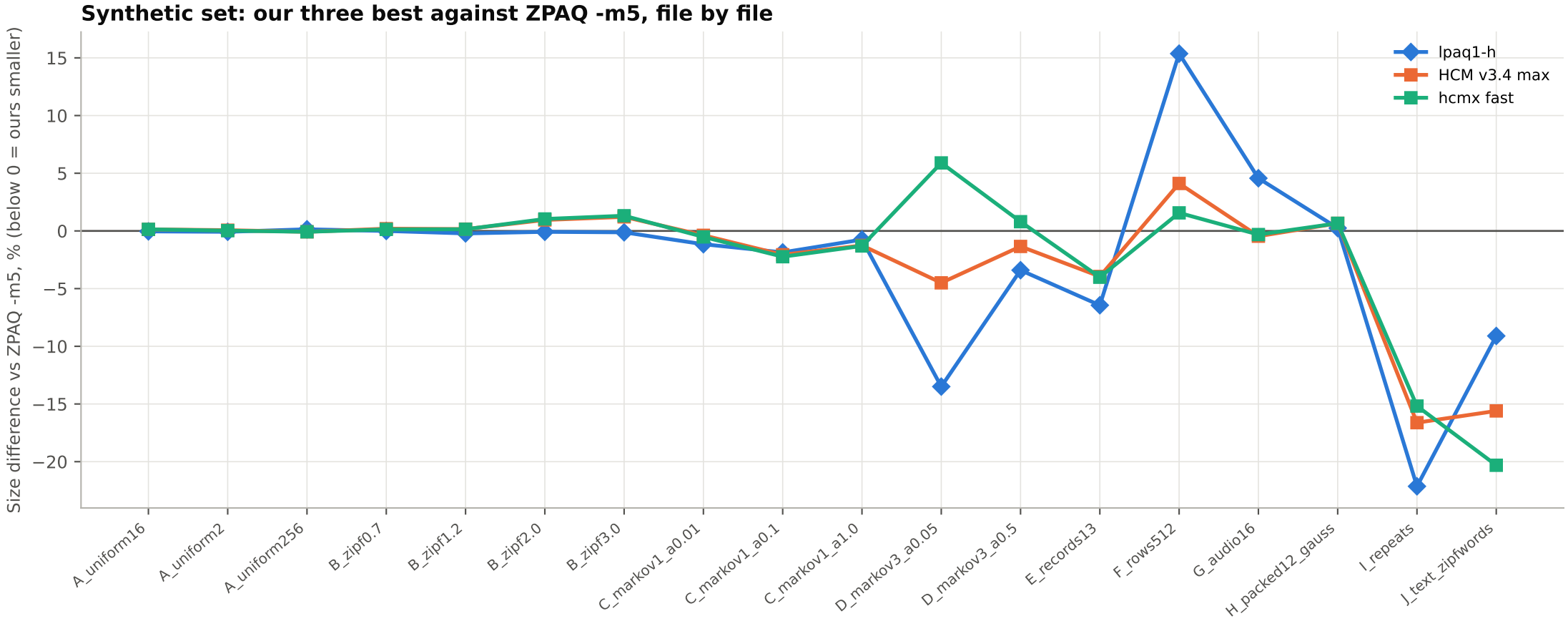

Synthetic set: size vs compression speed
Compressed size, % of original (lower is better)
Compression speed, MB/s (log)
lz4 -1
gzip -9
zstd -3
zstd -19
bzip2 -9
xz -9e
bzip2-hi
xz-hi
ZPAQ -m5
lpaq1-h
lpaq1 -6
hcmx fast
HCM v3.4 max
HCM v3.4 fast


Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line


Synthetic set, close-up of the smallest group: decode
Compressed size, % of original (lower is better)
51.00
50.00
49.00
48.00
47.00
46.00
45.00
xz-hi
lpaq1 -6
ZPAQ -m5
hcmx fast
HCM v3.4 fast
lpaq1-h
HCM v3.4 max
10^{-1}
10^{0}
10^{1}
Decompression speed, MB/s (log)
Synthetic set, close-up: compression
Compressed size, % of original (lower is better)
51.00
50.00
49.00
48.00
47.00
46.00
45.00
xz-hi
lpaq1 -6
ZPAQ -m5
hcmx fast
HCM v3.4 fast
lpaq1-h
HCM v3.4 max
10^{-2}
10^{-1}
10^{0}
Compression speed, MB/s (log)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line


Synthetic set: size vs peak memory
Size, % of original
85
80
75
70
65
60
55
50
45
lz4 -1
gzip -9
zstd -3
bzip2 -9
zstd -19
bzip2-hi
xz -9e
xz-hi
HCM v3.4 max
HCM v3.4 fast
hcmx fast
ZPAQ -m5
lpaq1 -6
lpaq1-h
10^1
10^2
Peak memory, MB (log; lower is lighter)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line


Synthetic set: file by file, three strongest standard tools (dashed) vs our three best (solid)
Size, % of original (lower is better)
100
80
60
40
20
ZPAQ -m5 (Standard) - total 8,201,855 B
lpaq1 -6 (Standard) - total 8,656,077 B
xz -9e (Standard) - total 9,342,648 B
lpaq1-h (Mutant) - total 8,113,745 B
HCM v3.4 max (Original) - total 8,127,874 B
hcmx fast (Original) - total 8,168,432 B
A_uniform16
A_uniform2
A_uniform256
B_zipf0.7
B_zipf1.2
B_zipf2.0
B_zipf3.0
C_markov1_a0.01
C_markov1_a0.1
C_markov1_a1.0
D_markov3_a0.05
D_markov3_a0.5
E_records13
F_rows512
G_audio16
H_packed12_gauss
I_repeats
J_text_zipfwords


Synthetic set: our three best against ZPAQ -m5, file by file
Size difference vs ZPAQ -m5, % (below 0 = ours smaller)
15
10
5
0
−5
−10
−15
−20
lpaq1-h
HCM v3.4 max
hcmx fast
A_uniform16
A_uniform2
A_uniform256
B_zipf0.7
B_zipf1.2
B_zipf2.0
B_zipf3.0
C_markov1_a0.01
C_markov1_a0.1
C_markov1_a1.0
D_markov3_a0.05
D_markov3_a0.5
E_records13
F_rows512
G_audio16
H_packed12_gauss
I_repeats
J_text_zipfwords